SQL Interview Questions with Solutions

1. What do DDL, DML, and DCL stand for?

Solution:

- **DDL — Data Definition Language:** deals with database schemas, as well as the description of how data resides in the database (e.g. CREATE, ALTER, DROP).
- **DML — Data Manipulation Language:** helps modify the content/data of a table (e.g. INSERT, UPDATE, DELETE).
- **DCL — Data Control Language:** controls access and permissions on database objects (e.g. GRANT, REVOKE).

2. How can you filter duplicate data while retrieving records from a table?

Solution:

A DISTINCT keyword is used to filter the duplicate data from the table while retrieving the records from a table.

```
SELECT DISTINCT city FROM customers;
```

3. What is the difference between CHAR and VARCHAR data types?

Solution:

When a table is created, CHAR is used to define the fixed length of the table's columns. The length value could be in the range of 1–255. The VARCHAR command is used to adjust the column and table lengths as required, so it only stores as much space as the actual value needs.

4. What is the difference between NOW() and CURRENT_DATE()?

Solution:

Both NOW() and CURRENT_DATE() are built-in MySQL functions. NOW() is used to show the current date and time of the server, and CURRENT_DATE() is used to show only the date of the server.

5. What are Joins and GROUP BY in SQL?

Solution:

The SQL statement that is used to make a connection between two or more tables based on matching columns is called a join. It is mainly used for complex queries.

The GROUP BY clause is a SQL command that is used to group rows that have the same values.

6. Explain the different types of MySQL joins.

Solution:

- **Inner Join:** the default join — returns records when the values match in the joining tables.
- **Left Outer Join:** returns all the records from the left table, plus the matched records from the right table.
- **Right Outer Join:** returns all the records from the right table, plus the matched records from the left table.
- **Full Outer Join:** returns all the records that match from the left or right table.

7. What are the different types of strings used in database columns in MySQL?

Solution:

In MySQL, the different types of strings that can be used for database columns are SET, BLOB, VARCHAR, TEXT, ENUM, and CHAR.

8. How can you change the name of an existing table using a SQL statement?

Solution:

```
RENAME TABLE table_name TO new_name;
```

9. What is the difference between the Primary Key and the Foreign Key?

Solution:

The primary key in MySQL is used to identify every row of a table in a unique manner. For one table, there is only one primary key.

A foreign key refers to a field or a collection of fields in a database record that uniquely identifies a key field of another database record in some other table — in simple terms, it establishes a link between records in two different tables in a database. A foreign key defined in a table refers to the primary key of some other table.

> Primary Key can't be null; Foreign Key accepts null values.

10. When do we use the LIKE operator in SQL?

Solution:

The LIKE operator is used in the WHERE clause of the SELECT, DELETE, and UPDATE statements to filter data based on patterns.

There are two wildcards often used in conjunction with the LIKE operator:

- **%** — represents zero, one, or multiple characters.
- **_** — represents a single character.

```
SELECT * FROM customers WHERE name LIKE 'R%';
```

11. What is the BLOB data type?

Solution:

BLOBs are binary large objects holding huge amounts of data.

12. How can you retrieve data in sorted order from a table?

Solution:

Using the ORDER BY clause, followed by ASC or DESC.

```
SELECT * FROM employees ORDER BY salary DESC;
```

13. What is a view? How can you create and drop a view in MySQL?

Solution:

In SQL, a view is a virtual table. A view contains rows and columns, just like a real table. The fields in a view are fields from one or more real tables in the database.

```
-- Create a view
CREATE VIEW active_customers AS
SELECT id, name, email FROM customers WHERE status = 'active';

-- Drop a view
DROP VIEW active_customers;
```

14. How can you retrieve a particular number of records from a table?

Solution:

LIMIT clause is used with the SQL statement to retrieve a particular number of records from a table. From which record and how many records will be retrieved are defined by the LIMIT clause.

```
SELECT * FROM orders LIMIT 10 OFFSET 20;
```

15. What is the difference between the WHERE and HAVING clauses in SQL?

Solution:

The main difference between the WHERE and HAVING clause comes when used together with GROUP BY — in that case, WHERE is used to filter rows before grouping, and HAVING is used to exclude records after grouping.

```
SELECT department, COUNT(*) AS headcount
FROM employees
WHERE status = 'active'
GROUP BY department
HAVING COUNT(*) > 5;
```

16. List out the Aggregate Functions in SQL.

Solution:

The ISO standard defines five aggregate functions:

- COUNT
- SUM
- AVG
- MIN
- MAX

17. What are the common SQL functions?

Solution:

- **CONCAT(A, B):** concatenates two string values to create a single string output. Often used to combine two or more fields into one.
- **FORMAT(X, D):** formats the number X to D significant digits.
- **CURDATE(), CURTIME():** returns the current date or time.
- **NOW():** returns the current date and time as one value.
- **MONTH(), DAY(), YEAR(), WEEK(), WEEKDAY():** extracts the given part of data from a date value.
- **HOUR(), MINUTE(), SECOND():** extracts the given part of data from a time value.
- **DATEDIFF(A, B):** determines the difference between two dates — commonly used to calculate age.
- **SUBTIME(A, B):** determines the difference between two times.

- **FROM_DAYS(INT):** converts an integer number of days into a date value.

18. What are SQL Constraints?

Solution:

A SQL CONSTRAINT is used to define rules to allow or restrict what values can be stored in columns. The purpose of inducing constraints is to enforce the integrity of a database.

- **NOT NULL:** ensures a column cannot store a null value.
- **UNIQUE:** ensures all values in a column are different.
- **PRIMARY KEY:** a combination of NOT NULL and UNIQUE — uniquely identifies each row.
- **FOREIGN KEY:** prevents actions that would break the link between two tables.
- **CHECK:** ensures all values in a column satisfy a specific condition.
- **DEFAULT:** sets a default value for a column when none is specified.

19. What is the default table in databases?

Solution:

The DUAL table is a special one row, one column table present by default in all Oracle databases.

```
SELECT current_date FROM dual;
```
